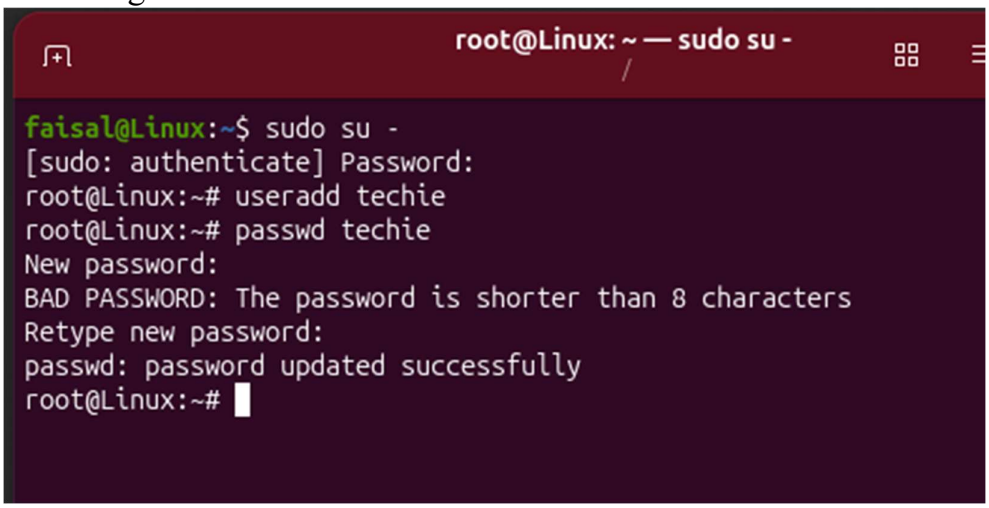
1. **Create user with techie and grant sudo permission**

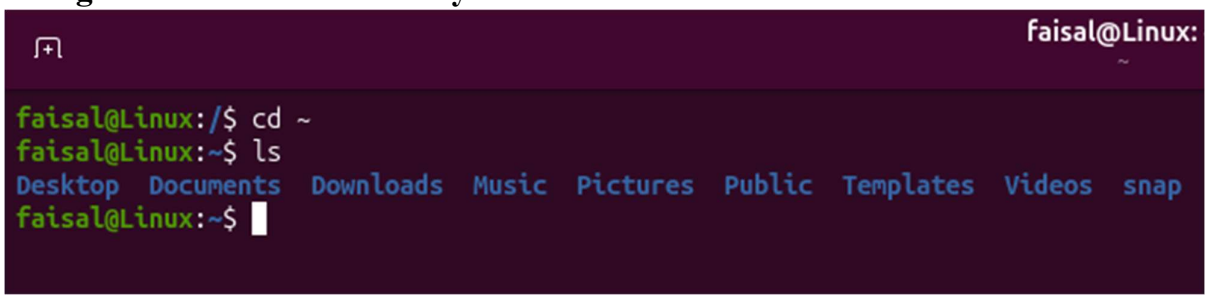
Switching to root user to create new user



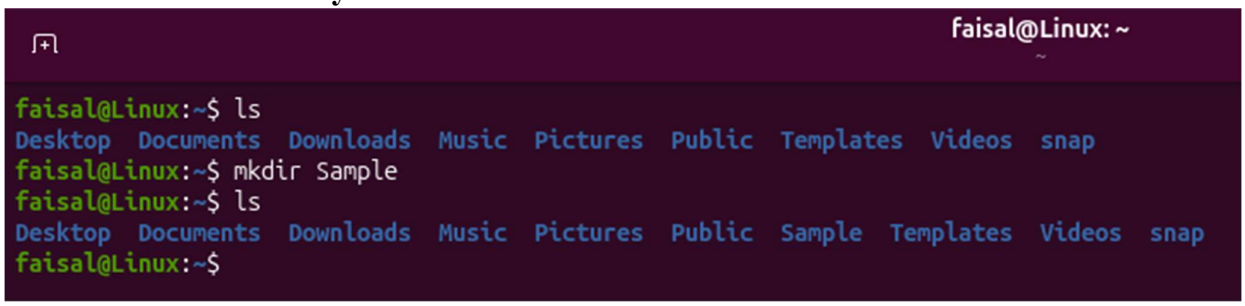
Granting the sudo access to the user



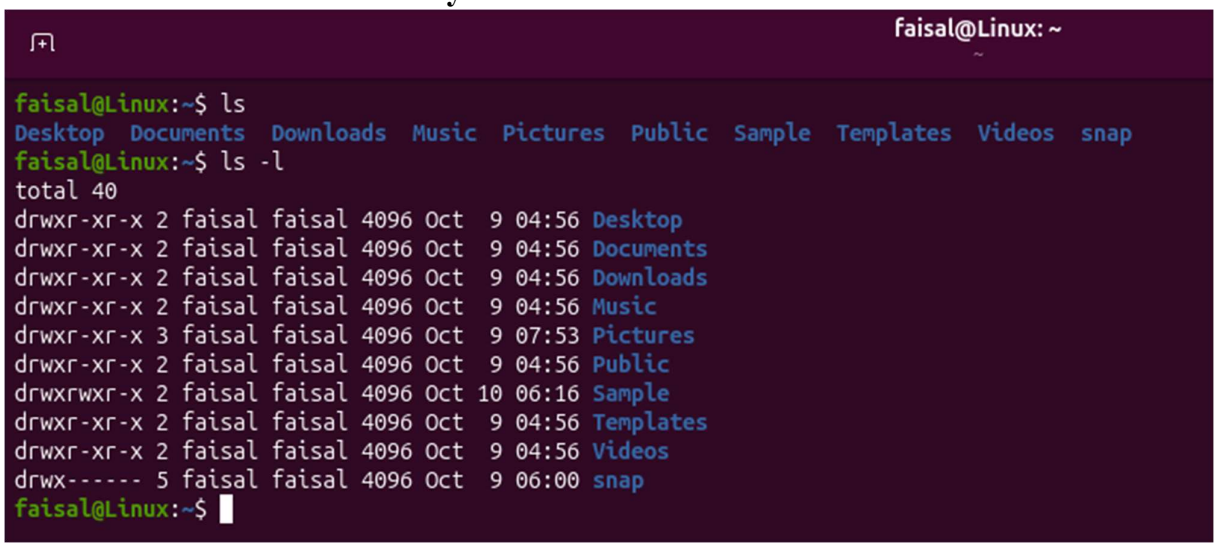
2. **Navigate to the home directory**



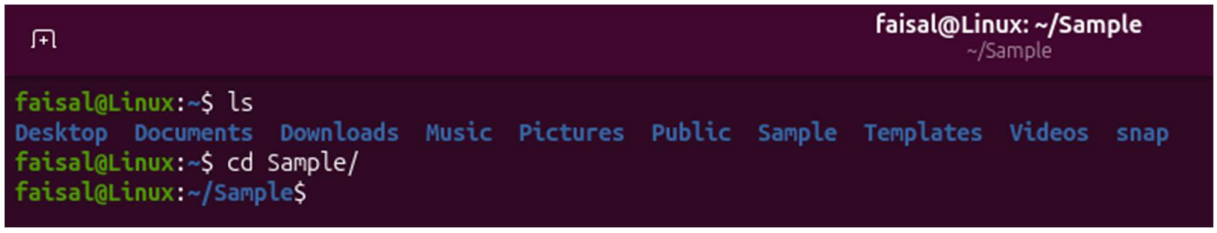
3. **Create a new directory**



4. **List the contents of a directory**



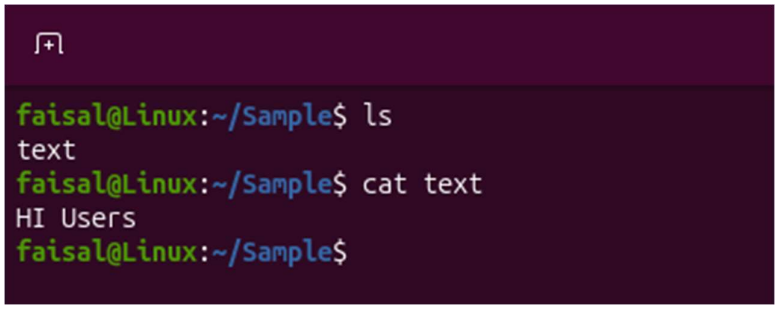
5. **Change the current directory**



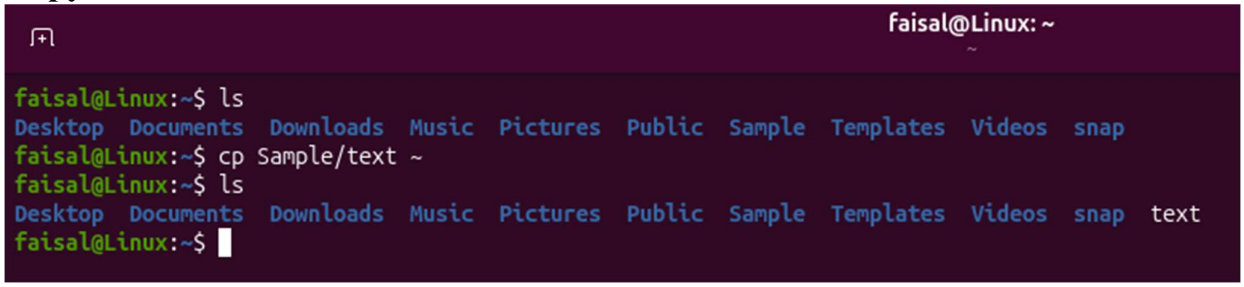
6. **Create a new empty file**



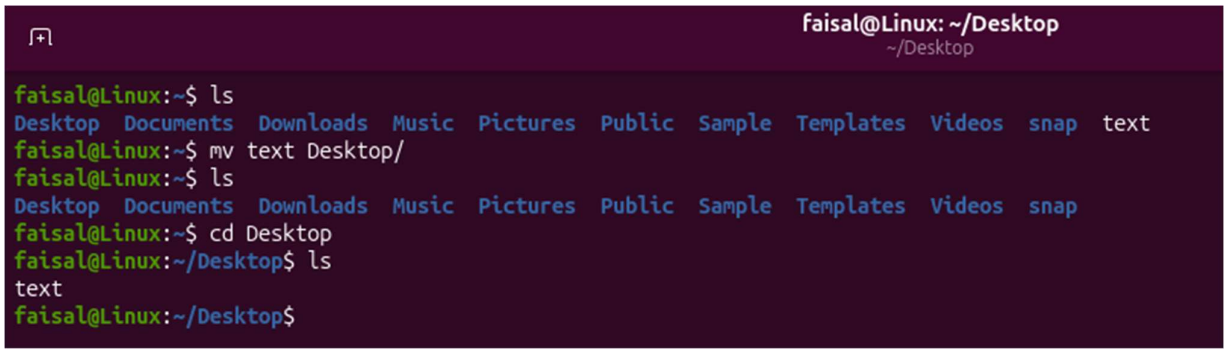
7. **View the contents of a file**



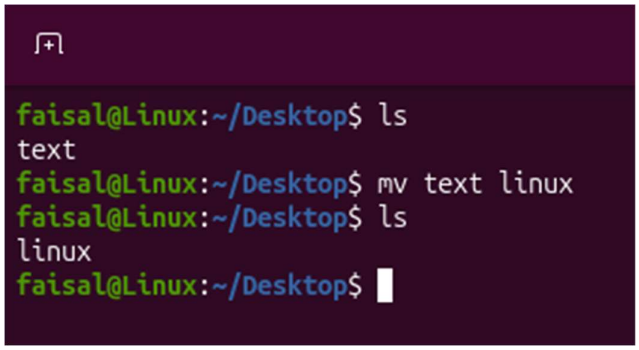
8. **Copy a file to another location**



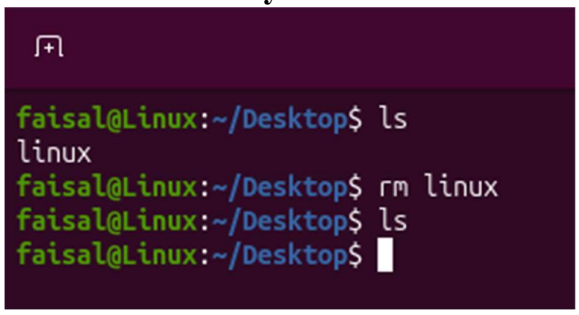
9. **Move the file to another location**



10. **Rename a file**

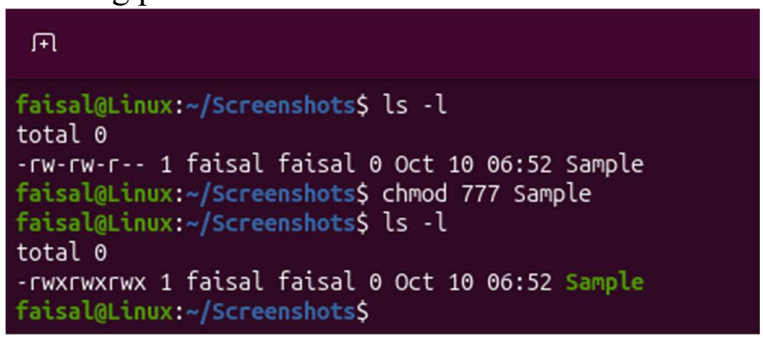


11. **Delete a directory**

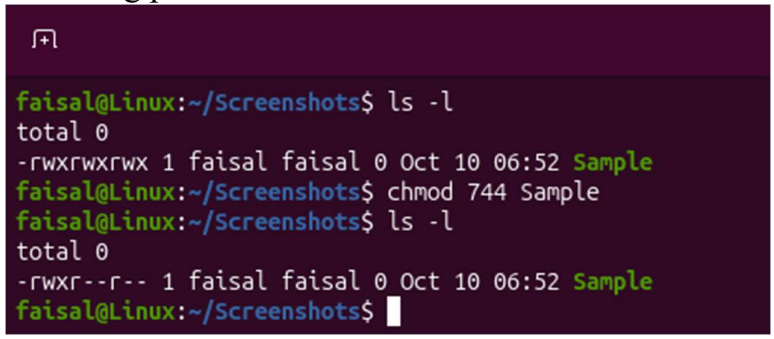


12. **Grant or revoke permission on a file or directory.**

Granting permission to all users



Revoking permission from other user



13. **View current date or time**

```
faisal@Linux:~/Screenshots$ date
Sat Oct 10 07:01:54 AM UTC 2026
faisal@Linux:~/Screenshots$
```

14. **Check system Uptime**

```
faisal@Linux:~$ uptime
 07:04:18 up  2:15,  3 users,  load average: 0.49, 0.39, 0.21
faisal@Linux:~$
```

15. **View the running processes**

faisal@Linux: ~/Sample
~/Sample

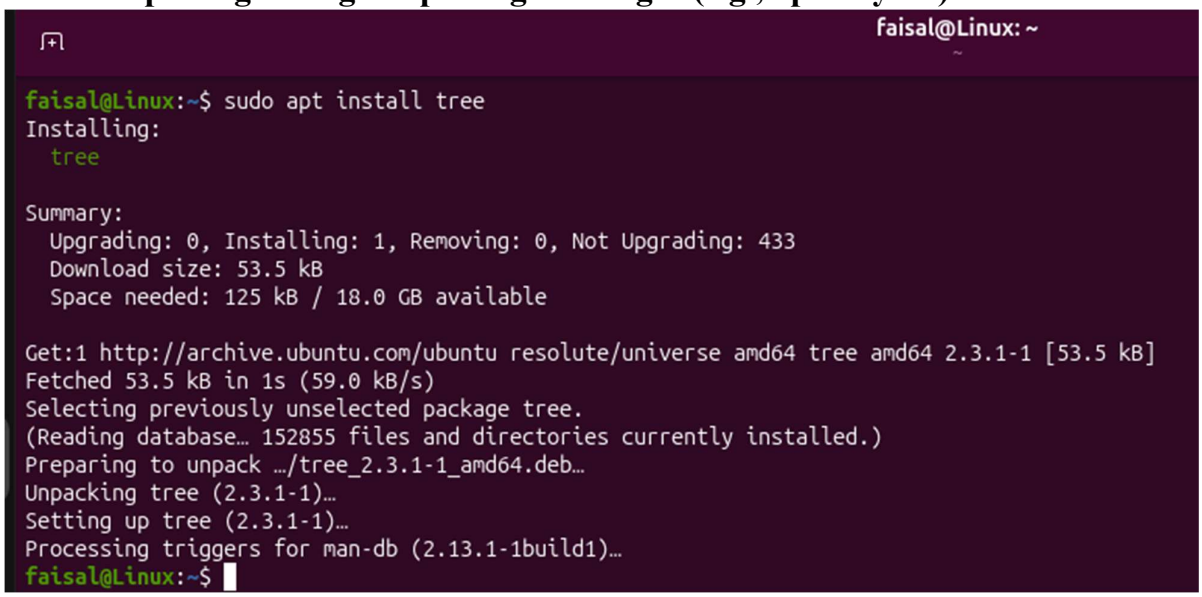
```
faisal@Linux:~/Sample$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 04:49 ?        00:00:37 /usr/lib/systemd/systemd --switched-root --system --d
root           2       0  0 04:49 ?        00:00:00 [kthreadd]
root           3       2  0 04:49 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 04:49 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 04:49 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 04:49 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 04:49 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 04:49 ?        00:00:00 [kworker/R-netns]
root          10       2  0 04:49 ?        00:00:03 [kworker/0:0H-kblockd]
root          13       2  0 04:49 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 04:49 ?        00:00:05 [ksoftirqd/0]
root          15       2  0 04:49 ?        00:00:11 [rcu_preempt]
root          16       2  0 04:49 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17       2  0 04:49 ?        00:00:09 [rcu_exp_gp_kthread_worker]
root          18       2  0 04:49 ?        00:00:01 [migration/0]
root          19       2  0 04:49 ?        00:00:00 [kprobe-optimizer]
root          20       2  0 04:49 ?        00:00:00 [idle_inject/0]
root          21       2  0 04:49 ?        00:00:00 [cpuhp/0]
root          22       2  0 04:49 ?        00:00:00 [cpuhp/1]
root          23       2  0 04:49 ?        00:00:00 [idle_inject/1]
root          24       2  0 04:49 ?        00:00:01 [migration/1]
root          25       2  0 04:49 ?        00:00:00 [ksoftirqd/1]
root          28       2  0 04:49 ?        00:00:00 [cpuhp/2]
root          29       2  0 04:49 ?        00:00:00 [idle_inject/2]
root          30       2  0 04:49 ?        00:00:01 [migration/2]
root          31       2  0 04:49 ?        00:00:00 [ksoftirqd/2]
root          33       2  0 04:49 ?        00:00:02 [kworker/2:0H-kblockd]
root          34       2  0 04:49 ?        00:00:00 [cpuhp/3]
root          35       2  0 04:49 ?        00:00:00 [idle_inject/3]
root          36       2  0 04:49 ?        00:00:01 [migration/3]
root          37       2  0 04:49 ?        00:00:00 [ksoftirqd/3]
root          40       2  0 04:49 ?        00:00:00 [cpuhp/4]
root          41       2  0 04:49 ?        00:00:00 [idle_inject/4]
root          42       2  0 04:49 ?        00:00:01 [migration/4]
root          43       2  0 04:49 ?        00:00:00 [ksoftirqd/4]
root          46       2  0 04:49 ?        00:00:00 [cpuhp/5]
root          47       2  0 04:49 ?        00:00:00 [idle_inject/5]
root          48       2  0 04:49 ?        00:00:00 [migration/5]
```

16. **Kill a running process**

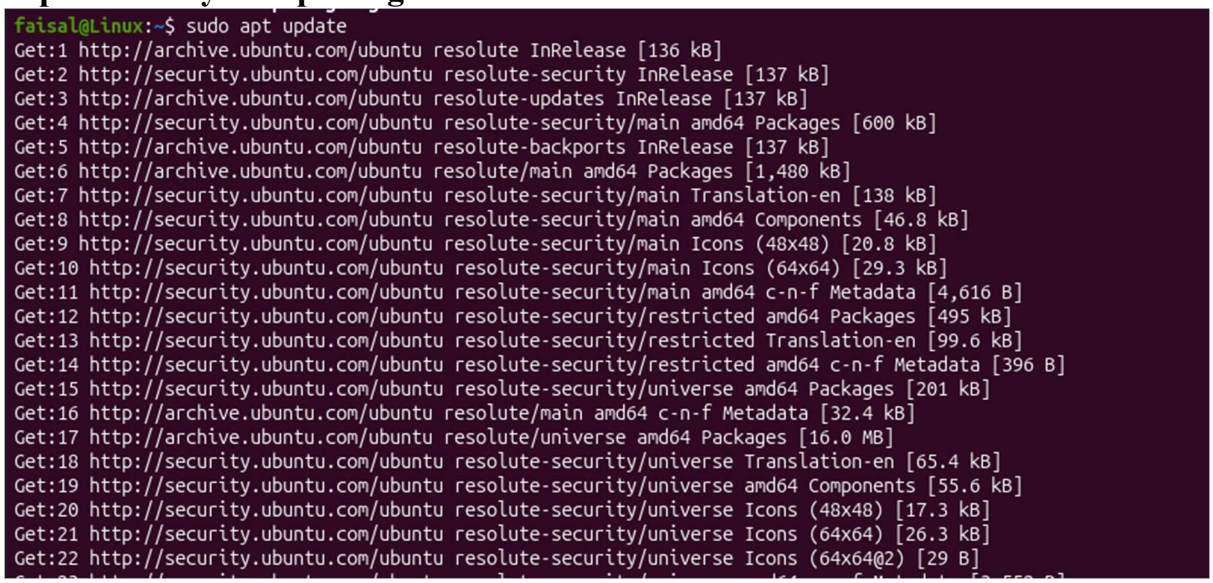
Using the pid of running process forcefully killing the process

```
faisal     16076   15627 99 07:27 pts/5    00:00:00 ps -ef
faisal@Linux:~/Sample$ kill -9 12886
faisal@Linux:~/Sample$
```

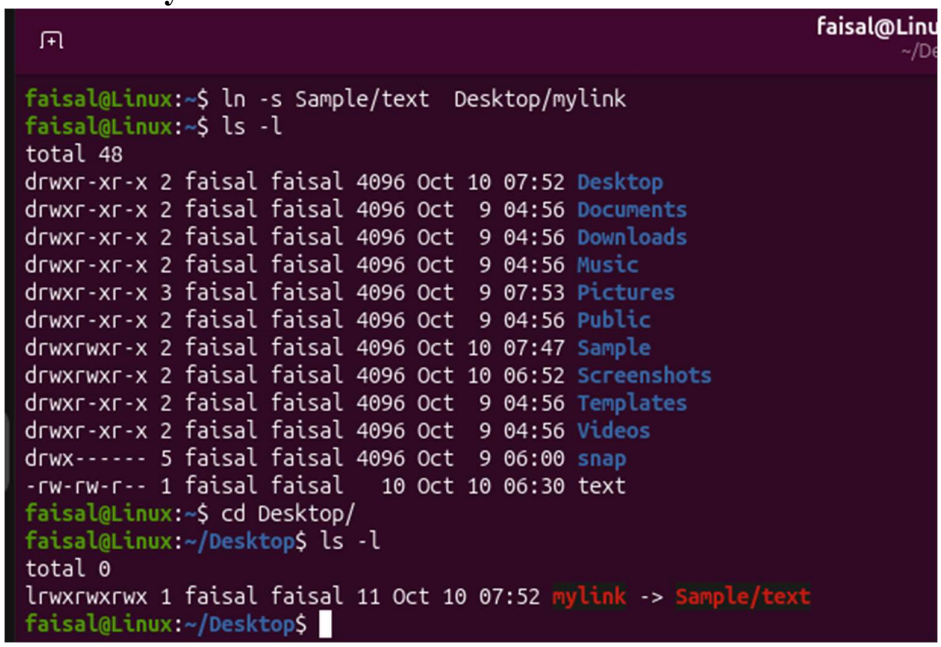
17. **Install a package using the package manager (e.g., apt or yum)**



18. **Update the system packages**



19. **Create a symbolic link**



20. Search for files using the find command

```
faisal@Linux:~$ sudo find / -name etc
find: ‘/run/user/1000/gvfs’: Permission denied
find: ‘/run/user/1000/doc’: Permission denied
/usr/share/factory/etc
/usr/local/etc
/etc
/etc/apparmor.d/tunables/etc
/var/lib/dracut/console-setup-dir/etc
/var/lib/ubuntu-advantage/apt-esm/etc
/snap/gnome-46-2404/153/etc
/snap/snapd/26865/etc
/snap/snapd/26865/usr/lib/snapd/apparmor.d/tunables/etc
/snap/core24/1587/etc
/snap/core24/1587/etc/apparmor.d/tunables/etc
/snap/bare/5/etc
find: ‘/proc/2062/task/2062/net’: Invalid argument
find: ‘/proc/2062/net’: Invalid argument
faisal@Linux:~$
```

21. Compress and decompress files using tar.

Compressing a folder

```
faisal@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Sample  Screenshots  Templates  Videos  snap  text
faisal@Linux:~$ tar cvf techie.tar Sample/
Sample/
Sample/.text.swp
faisal@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Sample  Screenshots  Templates  Videos  snap  techie.tar  text
faisal@Linux:~$
```

Decompressing the Tar file

```
faisal@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Sample  Screenshots  Templates  Videos  snap  techie.tar  text
faisal@Linux:~$ sudo tar xvf techie.tar
Sample/
Sample/.text.swp
faisal@Linux:~$
```

22. Monitor system resources with top or htop

```
faisal@Linux:/$ top
```

```
top - 05:21:54 up  1:37,  1 user,  load average: 1.03, 0.39, 0.18
Tasks: 325 total,   1 running, 324 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.7 us,  3.4 sy,  0.0 ni, 94.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3864.5 total,    194.7 free,   2272.7 used,   1716.1 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   1591.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2252 mysql     20   0  966096 527944  44704 S   7.1  13.3   3:47.02 mysqld
   8358 faisal    20   0   22096   5968   3812 R   7.1   0.2   0:00.05 top
      1 root      20   0   26224  17080  11724 S   0.0   0.4   0:28.37 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.60 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
```

23. Create and manage user groups

Creating user group

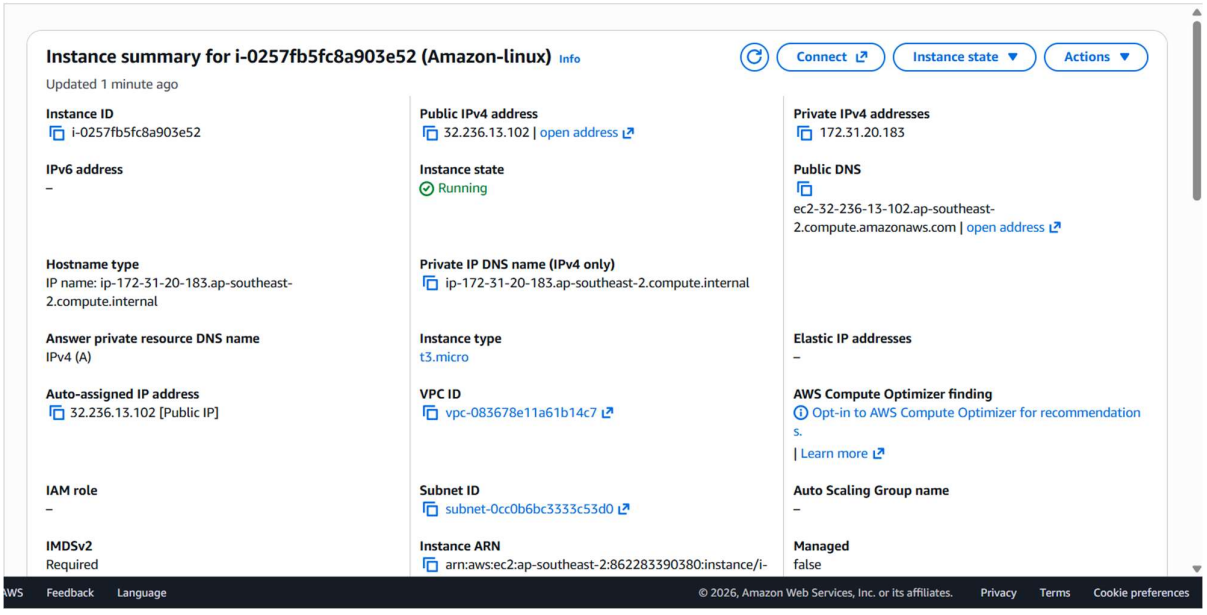
```
faisal@Linux:~$ sudo groupadd techie_horizon
[sudo: authenticate] Password:
faisal@Linux:~$ cat /etc/group | grep techie_horizon
techie_horizon:x:1002:
faisal@Linux:~$
```

Managing the user groups

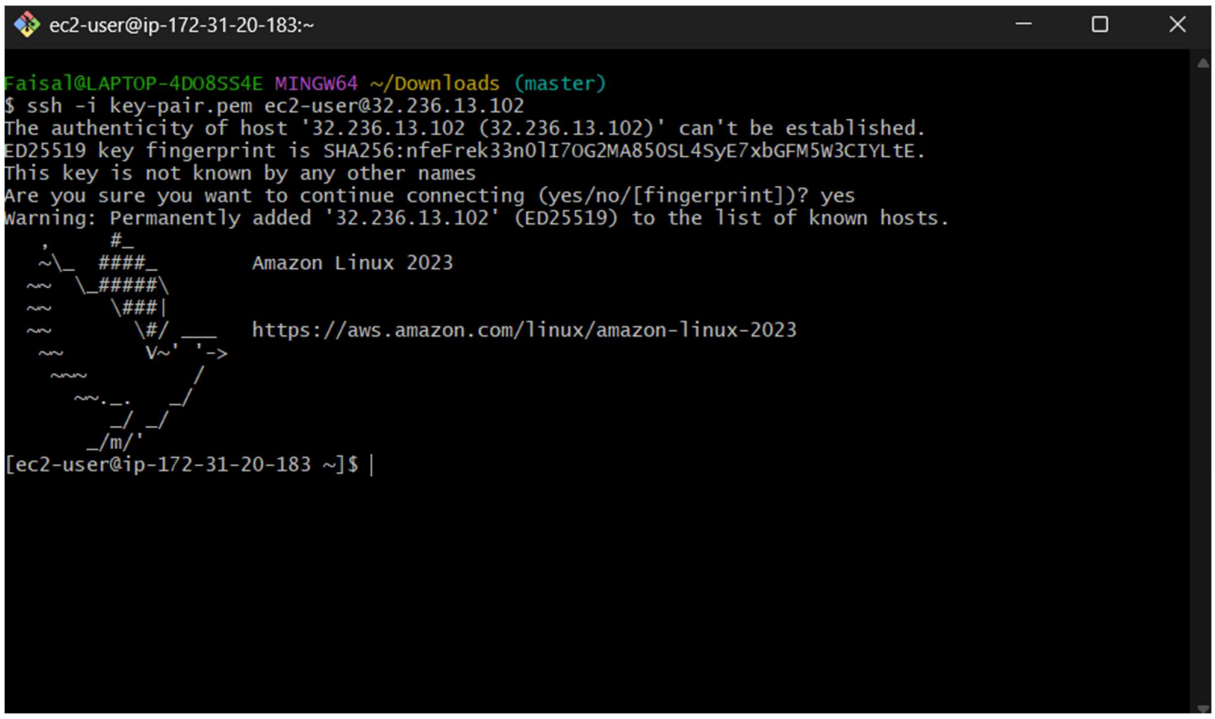
```
faisal@Linux:~$ sudo usermod -aG techie_horizon techie
faisal@Linux:~$ cat /etc/group | grep techie_horizon
techie_horizon:x:1002:techie
faisal@Linux:~$ sudo gpasswd -d techie techie_horizon
Removing user techie from group techie_horizon
faisal@Linux:~$ cat /etc/group | grep techie_horizon
techie_horizon:x:1002:
faisal@Linux:~$
```

24. Set up ssh password less authentication

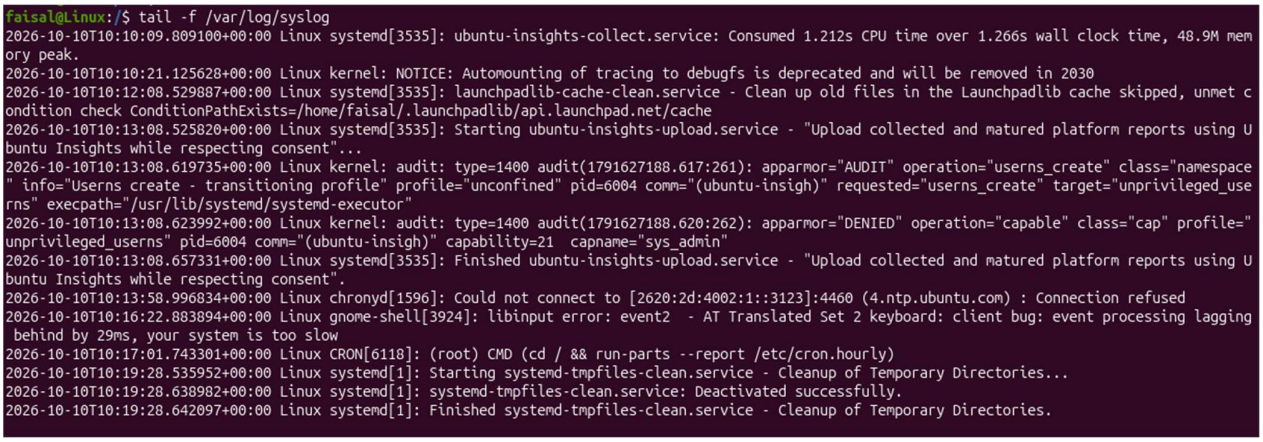
Create an ec2 instance



Open git bash anywhere and keep the key pair file in that directory



25. Monitor log files using tail or grep

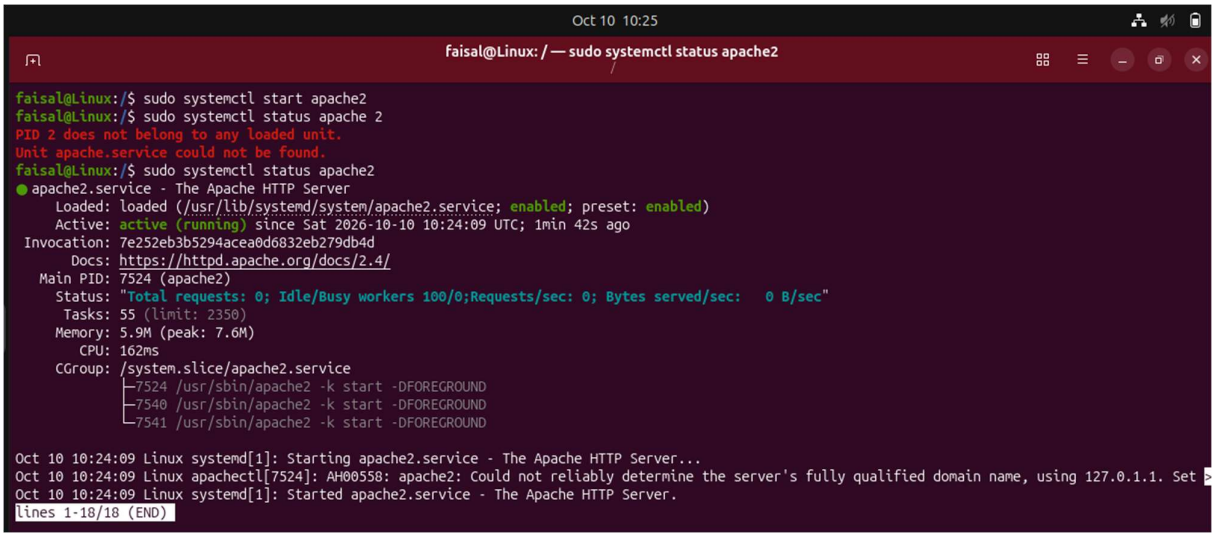


26. Set up a web server (e.g., Apache or ngnix)

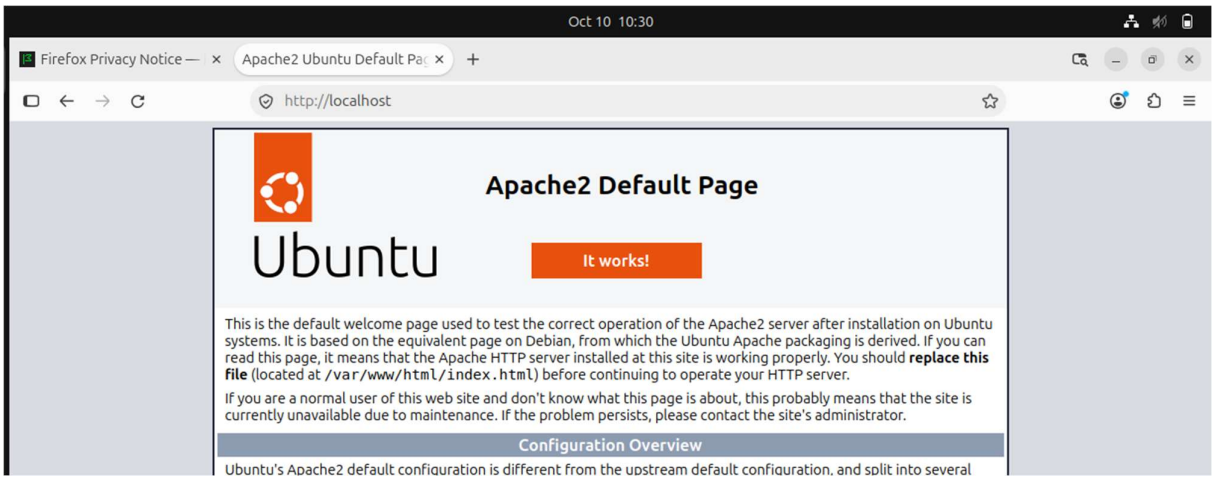
Installing apache package



Starting the package with systemctl and checking its status

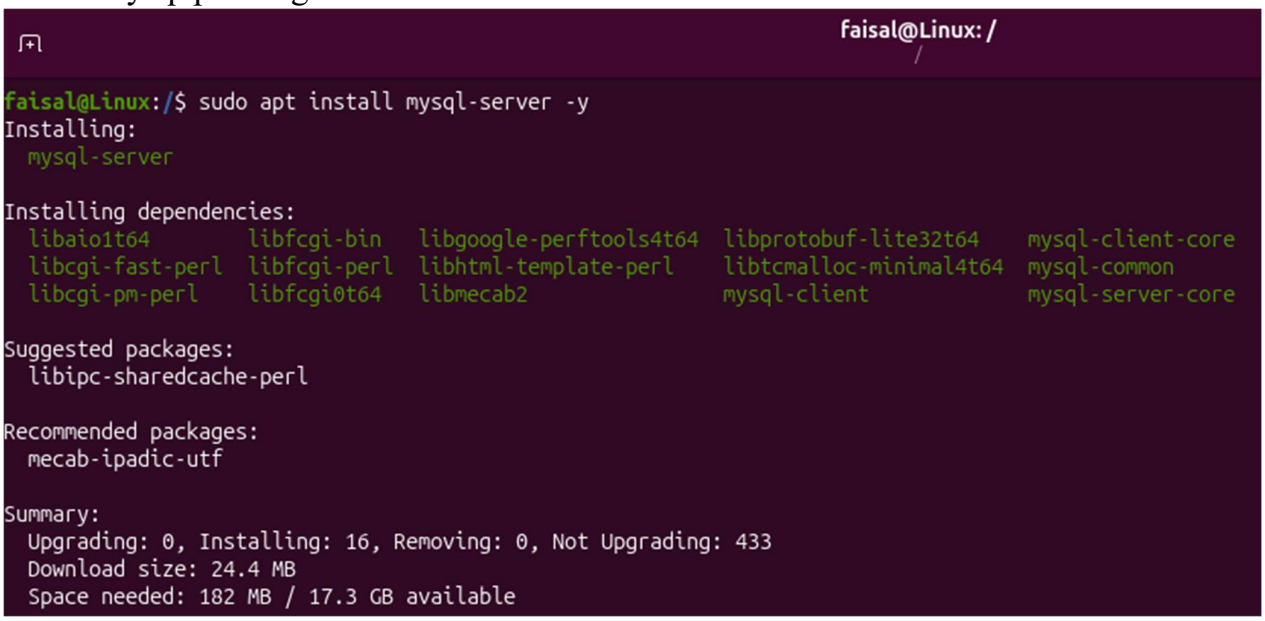


Checking the port in website



27. Configure and secure a MySQL Database

Install mysql pacakage



Starting the package with systemctl command and checking its status

```
faisal@Linux:/$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 10:37:36 UTC; 4min 9s ago
 Invocation: 82fa380b22b64b36a42a9914ef5dfbce
    Process: 10858 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS
   Main PID: 10867 (mysqld)
     Status: "Server is operational"
      Tasks: 34 (limit: 2350)
     Memory: 478.5M (peak: 479.4M)
        CPU: 10.082s
     CGroup: /system.slice/mysql.service
             └─10867 /usr/sbin/mysqld

Oct 10 10:37:34 Linux systemd[1]: Starting mysql.service - MySQL Community Server...
Oct 10 10:37:36 Linux systemd[1]: Started mysql.service - MySQL Community Server.
faisal@Linux:/$
```

Establishing connection and using the mysql terminal

```
faisal@Linux:/$ sudo mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.11-0ubuntu0.26.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

28. Set up a application server (e.g., Apache tomcat)

Installing tomcat10

```
faisal@Linux:/$ sudo apt install tomcat10 -y
Installing:
  tomcat10

Installing dependencies:
  ca-certificates-java  libbcel-java             libcommons-lang3-java                libeclipse-osgi-java     openjdk-25-jre-headless
  default-jre-headless  libcommons-codec-java    libcommons-parent-java               libosgi-annotation-java  tomcat-jakartaee-migration
  java-common           libcommons-compress-java libeclipse-jdt-core-compiler-batch-java libtcnative-2          tomcat10-common
  libapache-pom-java    libcommons-io-java       libeclipse-jdt-core-java             libtomcat10-java

Suggested packages:
  default-jre  libxz-java                 fonts-dejavu-extra    fonts-ipafont-mincho  | fonts-wqy-zenhei  tomcat10-admin  tomcat10-examples
  libasm-java  libosgi-annotation-java-doc fonts-ipafont-gothic fonts-wqy-microhei    fonts-indic        tomcat10-docs   tomcat10-user

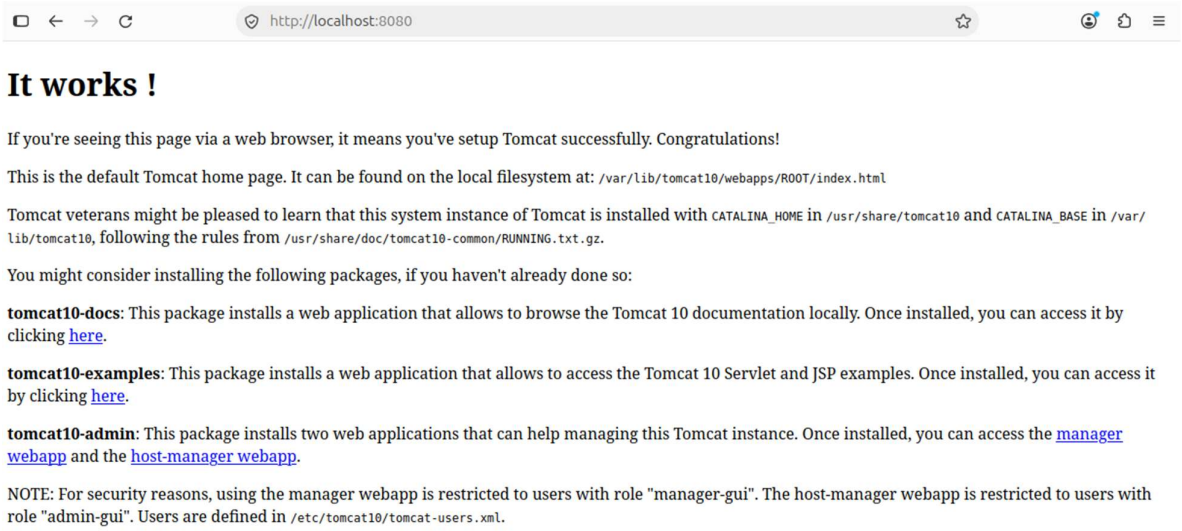
Summary:
  Upgrading: 0, Installing: 20, Removing: 0, Not Upgrading: 433
  Download size: 77.6 MB
  Space needed: 267 MB / 16.9 GB available
```

Starting the package with systemctl and checking it's status

```
faisal@Linux:/$ sudo systemctl start tomcat10
faisal@Linux:/$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 10:50:21 UTC; 3min 38s ago
 Invocation: 77fe4e0cf1504133aa884a24d44ab784
       Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
    Process: 12815 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 12820 (java)
      Tasks: 41 (limit: 2350)
     Memory: 149.6M (peak: 151M)
        CPU: 29.401s
     CGroup: /system.slice/tomcat10.service
             └─12820 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/logging.properties -Djava.util.logging

Oct 10 10:50:25 Linux tomcat10[12820]: OpenSSL successfully initialized [OpenSSL 3.5.5 27 Jan 2026]
Oct 10 10:50:27 Linux tomcat10[12820]: Initializing ProtocolHandler ["http-nio-8080"]
Oct 10 10:50:28 Linux tomcat10[12820]: Server initialization in [3720] milliseconds
Oct 10 10:50:30 Linux tomcat10[12820]: Starting service [Catalina]
Oct 10 10:50:30 Linux tomcat10[12820]: Starting Servlet engine: [Apache Tomcat/10.1.55 (Ubuntu)]
Oct 10 10:50:30 Linux tomcat10[12820]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
Oct 10 10:50:36 Linux tomcat10[12820]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a comple
Oct 10 10:50:36 Linux tomcat10[12820]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has finished in [6,007] ms
Oct 10 10:50:36 Linux tomcat10[12820]: Starting ProtocolHandler ["http-nio-8080"]
Oct 10 10:50:36 Linux tomcat10[12820]: Server startup in [7994] milliseconds
lines 1-23/23 (END)
```

Viewing the port to check



29. Create a service file for Apache Tomcat (Should execute by using systemctl command)

Checking for service file of apache tomcat

```
faisal@Linux:/$ systemctl cat tomcat10
# /usr/lib/systemd/system/tomcat10.service
#
# Systemd unit file for Apache Tomcat
#

[Unit]
Description=Apache Tomcat 10 Web Application Server
Documentation=https://tomcat.apache.org/tomcat-10.0-doc/index.html
After=network.target
RequiresMountsFor=/var/log/tomcat10 /var/lib/tomcat10

[Service]

# Configuration
Environment="CATALINA_HOME=/usr/share/tomcat10"
Environment="CATALINA_BASE=/var/lib/tomcat10"
Environment="CATALINA_TMPDIR=/tmp"
Environment="JAVA_OPTS=-Djava.awt.headless=true"

# Lifecycle
Type=simple
ExecStartPre=+/usr/libexec/tomcat10/tomcat-update-policy.sh
ExecStart=/bin/sh /usr/libexec/tomcat10/tomcat-start.sh
SuccessExitStatus=143
Restart=on-abort

# Logging
SyslogIdentifier=tomcat10

# Security
User=tomcat
Group=tomcat
PrivateTmp=yes
AmbientCapabilities=CAP_NET_BIND_SERVICE
NoNewPrivileges=true
CacheDirectory=tomcat10
CacheDirectoryMode=750
ProtectSystem=strict
ReadWritePaths=/etc/tomcat10/Catalina/
```

Running the service with systemctl

```
faisal@Linux:/$ sudo systemctl start tomcat10
faisal@Linux:/$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 10:50:21 UTC; 3min 38s ago
 Invocation: 77fe4e0cf1504133aa884a24d44ab784
       Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
    Process: 12815 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 12820 (java)
      Tasks: 41 (limit: 2350)
     Memory: 149.6M (peak: 151M)
        CPU: 29.401s
     CGroup: /system.slice/tomcat10.service
             └─12820 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/logging.properties -Djava.util.logging>

Oct 10 10:50:25 Linux tomcat10[12820]: OpenSSL successfully initialized [OpenSSL 3.5.5 27 Jan 2026]
Oct 10 10:50:27 Linux tomcat10[12820]: Initializing ProtocolHandler ["http-nio-8080"]
Oct 10 10:50:28 Linux tomcat10[12820]: Server initialization in [3720] milliseconds
Oct 10 10:50:30 Linux tomcat10[12820]: Starting service [Catalina]
Oct 10 10:50:30 Linux tomcat10[12820]: Starting Servlet engine: [Apache Tomcat/10.1.55 (Ubuntu)]
Oct 10 10:50:30 Linux tomcat10[12820]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
Oct 10 10:50:36 Linux tomcat10[12820]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a comple>
Oct 10 10:50:36 Linux tomcat10[12820]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has finished in [6,007] ms
Oct 10 10:50:36 Linux tomcat10[12820]: Starting ProtocolHandler ["http-nio-8080"]
Oct 10 10:50:36 Linux tomcat10[12820]: Server startup in [7994] milliseconds
lines 1-23/23 (END)
```

30. **Print specific column from a delimited file (csv)**

```
faisal@Linux:~/Sample$ ls
List.csv  Practice.txt  Students.csv  marks  mylink  result.csv  srt_result.csv  srt_students.csv  student.csv  techie.tar  text
faisal@Linux:~/Sample$ cut -d',' -f1,3 student.csv
Name,City
faisal,hyderabad
rahul,Nalgonda
sayeed,Hyderabad
faisal@Linux:~/Sample$
```

31. **Filter and print lines based on a specific pattern or condition**

```
faisal@Linux:~/Sample$ cat student.csv
Name,Age,City
faisal,25,hyderabad
rahul,23,Nalgonda
sayeed,29,Hyderabad
faisal@Linux:~/Sample$ grep 'Nalgonda' student.csv
rahul,23,Nalgonda
faisal@Linux:~/Sample$
```

32. **Calculate and print the average, sum or other statistics of a column**

```
faisal@Linux:~$ awk '{sum +=$1; count++} END {print "sum= ",sum;print "average= ",sum/count}' marks
sum=  60
average=  20
faisal@Linux:~$
```

33. **Perform string manipulation, such as extract substring or changing the case**

```
faisal@Linux:~/Sample$ echo 'MohammedFaisal' | cut -c9-14
Faisal
faisal@Linux:~/Sample$ echo 'Hello User' | tr '[:lower:]' '[:upper:]'
HELLO USER
faisal@Linux:~/Sample$ echo 'Hello User' | tr '[:upper:]' '[:lower:]'
hello user
faisal@Linux:~/Sample$ echo 'Hello User' | tr 'a-z' 'A-Z'
HELLO USER
faisal@Linux:~/Sample$ echo 'Hello User' | tr 'A-Z' 'a-z'
hello user
faisal@Linux:~/Sample$ _
```

34. **Count the occurrences of specific pattern in a file**

```
faisal@Linux:~$ grep -c 'linux' text
4
faisal@Linux:~$ grep -o 'linux' text | wc -l
4
```

35. **Sort lines based on a specific field or column**

```
faisal@Linux:~$ cat student.csv
Name,Age,City
faisal,25,hyderabad
rahul,23,Nalgonda
sayeed,29,Hyderabad
faisal@Linux:~$ sort -t',' -k2,2n student.csv
Name,Age,City
rahul,23,Nalgonda
faisal,25,hyderabad
sayeed,29,Hyderabad
faisal@Linux:~$
```

36. Merge multiple files based on a common field or column

Sorting both files then using join to merge them

```
faisal@Linux:~$ cat Students.csv
id,  name, age, city
1, faisal, 18, NGKL
2, sayeed, 19, HYD
3, Junaid, 18, HYD

faisal@Linux:~$ cat result.csv
id, name, marks
3, juniad, 30/45
1, faisal, 39/45
2, sayeed, 42/45
faisal@Linux:~$ sort -k1,1 Students.csv > srt_students.csv
faisal@Linux:~$ sort -k1,1 result.csv > srt_result.csv
faisal@Linux:~$ join srt_students.csv srt_result.csv
1, faisal, 18, NGKL faisal, 39/45
2, sayeed, 19, HYD sayeed, 42/45
3, Junaid, 18, HYD juniad, 30/45
id, name, age, city  name, marks
faisal@Linux:~$
```

37. Substitute text in a file using search and replace

```
faisal@Linux:~$ cat Students.csv
id,  name, age, city
1, Faisal, 18, NGKL
2, sayeed, 19, HYD
3, Junaid, 18, HYD

faisal@Linux:~$ sed -i 's/Junaid/sami/g' Students.csv
faisal@Linux:~$ cat Students.csv
id,  name, age, city
1, Faisal, 18, NGKL
2, sayeed, 19, HYD
3, sami, 18, HYD

faisal@Linux:~$
```

38. Delete specific line base on a pattern or line number

Using sed to delete with specific condition

```
faisal@Linux:~$ cat List.csv
faisal
saboor
rahman
tabrez
rahul
faisal@Linux:~$ sed -i '/tabrez/d' List.csv
faisal@Linux:~$ cat List.csv
faisal
saboor
rahman
rahul
faisal@Linux:~$
```

Using sed to delete by line

```
faisal@Linux:~$ cat List.csv
faisal
saboor
rahman
rahul
faisal@Linux:~$ sed -i '4d' List.csv
faisal@Linux:~$ cat List.csv
faisal
saboor
rahman
faisal@Linux:~$
```

39. Append or insert text before or after a specific pattern or line

Inserting or appending line With sed

```
faisal@Linux:~$ cat List.csv
faisal
saboor
rahman
faisal@Linux:~$ sed -i '3i\Z' List.csv
faisal@Linux:~$ cat List.csv
faisal
saboor
Z
rahman
faisal@Linux:~$ sed -i '3a\Y' List.csv
faisal@Linux:~$ cat List.csv
faisal
saboor
Z
Y
rahman
faisal@Linux:~$
```

Inserting or appending text using sed

```
faisal@Linux:~$ cat List.csv
faisal
saboor
Z
Y
rahman
faisal@Linux:~$ sed -i '4s/$/ack/' List.csv
faisal@Linux:~$ cat List.csv
faisal
saboor
Z
Yack
rahman
faisal@Linux:~$ sed -i '3s/^/BUZ/' List.csv
faisal@Linux:~$ cat List.csv
faisal
saboor
BUZZ
Yack
rahman
faisal@Linux:~$
```

40. Print only specific lines from a file

```
faisal@Linux:~$ sed -n '2p' Practice.txt
Ubuntu is popular among Devs
faisal@Linux:~$ awk 'NR==2' Practice.txt
Ubuntu is popular among Devs
faisal@Linux:~$
```

41. Copy file from Linux to Windows Machine

Create file in virtual server to transfer

```
MINGW64:/c/Users/SUBHAN/Downloads

Faisal@LAPTOP-4DO8SS4E MINGW64 ~/Downloads (master)
$ ssh -i key-pair.pem ec2-user@32.236.13.102
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Oct 11 14:50:40 2026 from 106.192.33.77
[ec2-user@ip-172-31-20-183 ~]$ ls
amazon.txt
[ec2-user@ip-172-31-20-183 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-20-183 ~]$ exit
logout
Connection to 32.236.13.102 closed.
```

Then use scp to transfer from virtual server to windows

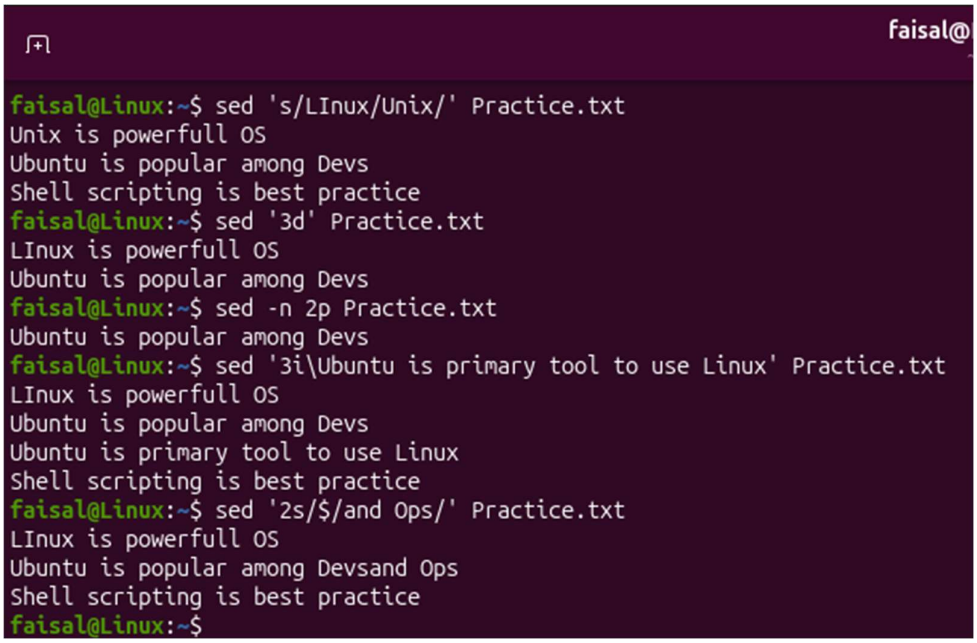
```
Faisal@LAPTOP-4DO8SS4E MINGW64 ~/Downloads (master)
$ scp -i key-pair.pem ec2-user@32.236.13.102:/home/ec2-user/amazon.txt ~/Downloads/
amazon.txt                                    100%   33     0.1KB/s   00:00
```

42. 5 use cases of AWK and 5 use cases on SED

Use cases of AWK

```
faisal@Linux:~$ awk 'NR==3' Practice.txt
Shell scripting is best practice
faisal@Linux:~$ awk 'NR>=2 && NR<=3' Practice.txt
Ubuntu is popular among Devs
Shell scripting is best practice
faisal@Linux:~$ awk '/Shell/' Practice.txt
Shell scripting is best practice
faisal@Linux:~$ awk '{print $1}' Practice.txt
LInux
Ubuntu
Shell
faisal@Linux:~$ awk 'END {print NR}' Practice.txt
3
faisal@Linux:~$
```

Use cases of SED


faisal@
faisal@Linux:~$ sed 's/LInux/Unix/' Practice.txt
Unix is powerfull OS
Ubuntu is popular among Devs
Shell scripting is best practice
faisal@Linux:~$ sed '3d' Practice.txt
LInux is powerfull OS
Ubuntu is popular among Devs
faisal@Linux:~$ sed -n 2p Practice.txt
Ubuntu is popular among Devs
faisal@Linux:~$ sed '3i\Ubuntu is primary tool to use Linux' Practice.txt
LInux is powerfull OS
Ubuntu is popular among Devs
Ubuntu is primary tool to use Linux
Shell scripting is best practice
faisal@Linux:~$ sed '2s/$/and Ops/' Practice.txt
LInux is powerfull OS
Ubuntu is popular among Devsand Ops
Shell scripting is best practice
faisal@Linux:~$